

# **S403011 Machine Learning Project: Weather Prediction**

## **Team: GWML25**

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## **1 Introduction**

Accurate short-term temperature forecasting is a fundamental task in meteorology, with important applications in energy management, agriculture, transportation, and environmental monitoring. While large-scale numerical weather prediction models form the backbone of operational forecasting, data-driven approaches based on historical observations provide a valuable complementary perspective, particularly for local, station-level predictions over short forecast horizons.

The goal of this project is to build machine learning models that predict the 2 m air temperature in Bern (`tre200h0`) at three forecast horizons: 12, 24, and 48 hours ahead, using hourly meteorological observations from 10 weather stations across Switzerland. Model development and evaluation are therefore conducted consistently for all three horizons.

The dataset consists of hourly meteorological measurements, including temperature, humidity, radiation, precipitation, and temporal variables, collected from multiple stations. These variables exhibit complex dependencies driven by atmospheric dynamics, diurnal cycles, seasonal effects, and spatial interactions between locations. As a result, the forecasting task poses challenges that range from capturing simple linear trends to modelling non-linear interactions and regime-dependent behaviour.

Rather than relying on a single modelling approach, the analysis follows a structured progression from simple to more flexible methods. Linear Regression is first employed as a transparent baseline model, providing interpretability and a reference level of performance. Next, K-Nearest Neighbours (KNN) is considered as a non-parametric approach capable of capturing local non-linear patterns in the data without assuming a global functional form. Finally, boosting-based models are applied to learn complex non-linear relationships and feature interactions through ensembles of decision trees, ultimately achieving the best predictive performance in this study.

A key component of the modelling pipeline is feature engineering. In addition to the original meteorological variables, derived features are constructed to encode temporal structure and spatial information, including cyclic representations of hour and season, lag-based temperature differences, and aggregated statistics across stations. While a broader set of engineered features is explored during development, the final models retain only a small subset of three additional features that consistently improve predictive accuracy without introducing unnecessary complexity.

Model performance is evaluated using the **Mean Absolute Error (MAE)** on held-out test data. Beyond aggregate error metrics, diagnostic plots and residual analyses are used to assess model's behaviour, identify systematic errors, and examine how forecast uncertainty evolves across horizons.

The results demonstrate that boosting-based models provide the most accurate and robust predictions for the 12-hour, 24-hour, and 48-hour temperature forecast in Bern, substantially outperforming both

linear regression and KNN. While simpler models remain valuable for benchmarking and interpretability, the findings highlight the importance of non-linear ensemble methods for capturing the complex structure of local temperature dynamics.

## 2 Exploratory Data Analysis

### 2.1 Dataset

The training dataset consists of hourly meteorological observations collected from 10 weather stations across Switzerland: Basel (BAS), Genève (GVE), Interlaken (INT), Sion (SIO), St. Gallen (STG), Davos (DAV), Zermatt (ZER), Andermatt (ANT), La Dôle (DOL), and Lugano (LUG). The training set contains 7,579 rows and 92 columns. The test dataset contains 3,247 rows and corresponds to observations from Bern.

Variables are organised by meteorological quantity and station identifier using the pattern `<variable>_<station>`. The dataset includes the following meteorological measurements (hourly aggregates unless stated otherwise):

- `fk1010h0`: wind speed scalar (hourly mean, m/s),
- `fk1010h3`: gust peak (3 seconds; hourly maximum, m/s),
- `gre000h0`: global radiation (hourly mean),
- `pp0qffh0`: atmospheric pressure reduced to sea level (QFF; hourly mean),
- `prestah0`: atmospheric pressure at barometric altitude (QFE; hourly mean),
- `rre150h0`: precipitation (hourly total),
- `sre000h0`: sunshine duration (hourly total),
- `tre200h0`: air temperature at 2 m above ground (hourly mean),
- `tre200h0_lag24h`: air temperature at 2 m above ground lagged by 24 hours,
- `ure200h0`: relative air humidity at 2 m above ground (hourly mean).

In addition to meteorological variables, the dataset contains temporal indicators `hour` and `season`. Most variables are stored as numeric types, while `season` is stored as an object/categorical field. We note that not all stations provide all variables (for example, `pp0qffh0` is not available for every location), which explains slight structural differences across station-specific feature groups.

## 2.2 Duplicate Observations

To verify the uniqueness of the observations, duplicate rows in the training dataset were identified using a row-wise comparison across all features. The duplicate check returned zero duplicated entries.

This confirms that each row in the dataset represents a unique hourly observation, and no further action related to duplicate removal was required.

## 2.3 Missing Data Analysis

Missing values were first quantified at the feature level using `isnull().sum()`, which shows that missingness is limited to a small number of entries across several variables. In total, 150 rows contain at least one missing value, corresponding to approximately 1.98% of the training set.

Figure 2.1 reports the percentage of missing values per feature (restricted to features with non-zero missingness). The plot confirms that missingness remains low across all affected variables, with no feature exhibiting a substantial proportion of missing observations.

To assess whether missingness occurs systematically across variables, we calculated the correlations between missing-value indicators and visualized them as a heatmap (Figure 2.2). The near-zero off-diagonal values indicate that missingness does not tend to co-occur across specific groups of variables, suggesting that missingness is largely unstructured.

Given both the low proportion of missing values and the absence of strong missingness dependencies, rows containing missing observations were removed from the training dataset rather than imputation.

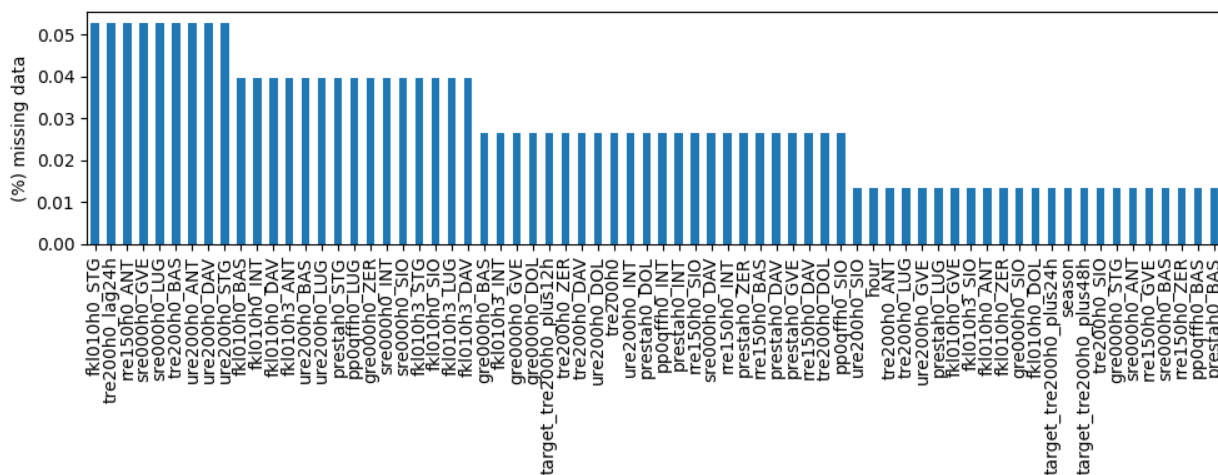


Figure 2.1: Percentage of missing values per feature in the training dataset (features with zero missingness omitted). Missingness remains low across all affected variables.

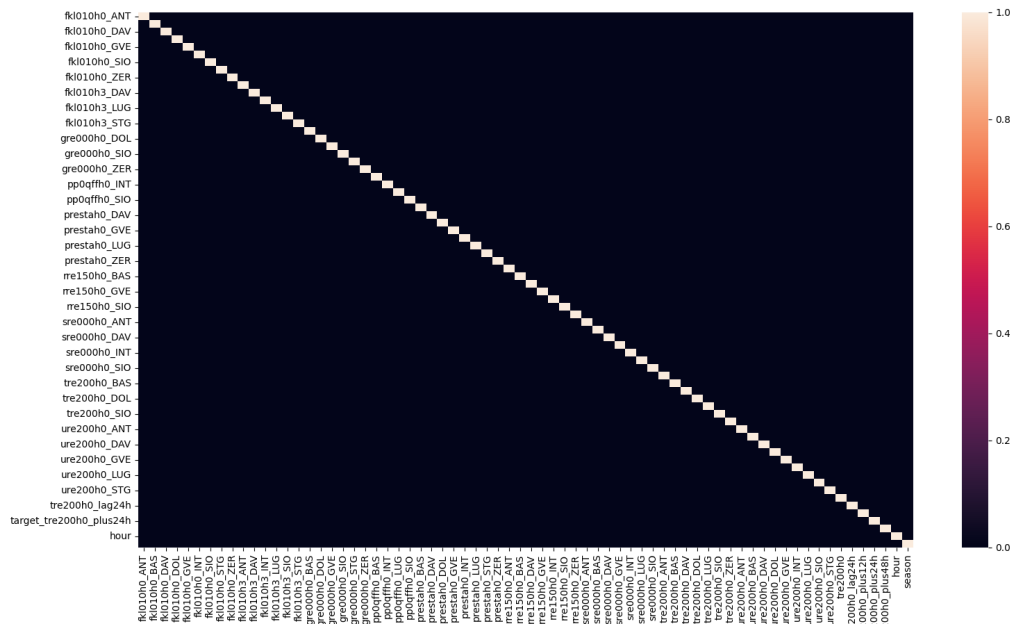


Figure 2.2: Correlation heatmap of missing-value indicators. Near-zero correlations suggest that missingness does not systematically co-occur across variables.

## 2.4 Minimal Preprocessing for Exploratory Analysis

Based on the results of the missing data analysis, rows containing missing observations were removed from the training dataset. After this step, the dataset contains 7,429 observations.

In addition, the variable `season` was converted from a generic object type to a categorical data type. This change reflects the discrete nature of the variable and ensures correct handling in summary statistics and visualizations.

No further preprocessing or feature transformations were applied at this stage, as the goal of this step is solely to support exploratory analysis rather than model optimization.

## 2.5 Data Distribution

The empirical distributions of the temporal variables `season` and `hour` were examined to assess data coverage across different time periods. All seasons are represented in the dataset; however, the number of observations is not perfectly balanced, with moderate variation in sample size between seasons. Similarly, observations are available for all 24 hours of the day, but the number of samples per hour is not uniform. This suggests that the dataset does not contain complete daily cycles for all days and that observations may be irregularly sampled or partially aggregated over time.

The non-uniform distributions of `season` and `hour` imply that these temporal variables should be interpreted as indicators rather than as part of a strictly ordered time series. This observation motivates treating `hour` and `season` as ordinal and potentially cyclical features in subsequent modeling steps, rather than assuming continuous temporal dependence.

## 2.6 Relationships Between Predictors and the Target

To explore relationships between meteorological predictors and the target variables, pairwise scatter plots were examined for groups of features corresponding to different physical quantities (e.g. temperature, wind, radiation, precipitation, pressure, and humidity), both across stations and with respect to the prediction targets.

Temperature-related variables exhibit a clear and strong linear relationship with the target variables. This pattern is consistent across stations and is expected given the physical continuity of air temperature over short forecasting horizons. The lagged temperature variable (`tre200h0_lag24h`) shows a similarly strong association with the targets.

In contrast, other meteorological variables such as wind speed, gust peak, precipitation, and sunshine duration show no clear linear relationship with the targets in the pairwise plots. These features display substantial dispersion and skewness, suggesting that their influence on the target may be indirect or require non-linear modeling or feature engineering to be captured effectively.

Atmospheric pressure and relative humidity show moderate structure in their own distributions but do not exhibit strong direct relationships with the target variables.

Overall, the exploratory analysis indicates that temperature is the dominant predictor in its raw form, while other variables may require transformation or interaction terms to contribute meaningfully to predictive performance.

## 2.7 Correlation Structure

Pairwise correlations among numerical features were examined to identify structural dependencies and potential multicollinearity. The analysis reveals strong correlations between identical meteorological variables measured at different weather stations, reflecting the spatial coherence of atmospheric conditions across Switzerland.

High correlations are particularly evident among temperature-related variables and atmospheric pressure measurements. These block-like patterns indicate substantial multicollinearity within specific groups of physically related features.

In contrast, variables such as wind speed, gust peak, precipitation, and sunshine duration exhibit weaker correlations with most other predictors. Relative humidity shows moderate negative correlations with temperature-related variables, consistent with known physical relationships.

Overall, the correlation structure suggests that while many features are strongly correlated within groups, different categories of meteorological variables provide complementary information. This motivates the use of regularization techniques or models robust to correlated inputs in subsequent modeling stages.

## 2.8 Outliers Analysis

An outlier analysis was conducted to assess the presence and structure of extreme values across the meteorological variables. Given the physical nature of the data, particular care was taken to distinguish between implausible measurement errors and legitimate extreme weather events.

Boxplots stratified by `season` and `hour` reveal the presence of numerous extreme values in several variables, especially wind-related features (wind speed and gust peaks), precipitation, sunshine duration, and global radiation. These outliers were not uniformly distributed across time: for example, wind speed exhibits higher variability during daytime hours, while precipitation and sunshine show strong concentration near zero with occasional extreme spikes.

To quantify outliers more systematically, the interquartile range (IQR) criterion was applied to all numerical features. A non-negligible number of observations fall outside the  $[Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR]$  range for many variables, particularly those known to be naturally skewed. Importantly, these extreme values occur consistently across stations and time periods, suggesting that they correspond to genuine meteorological phenomena rather than data quality issues.

Due to their physical interpretability and potential predictive relevance, outliers were not removed at this stage. Instead, the presence of skewed distributions motivated consideration of robust preprocessing strategies in the modelling phase. While logarithmic transformations are commonly applied to strictly positive, right-skewed variables, they were not adopted in this study because several features exhibit left-skewed or mixed skewness, for which log transformations are inappropriate.

Consequently, `RobustScaler` was selected as the primary scaling method. Robust scaling relies on median and interquartile range statistics and therefore reduces the influence of extreme values without removing them. This approach preserves informative extremes while mitigating their undue impact on model estimation, particularly for distance-based and linear models.

## 2.9 Temporal Structure and Cyclical Features

Although the dataset contains temporal indicators such as `hour` and `season`, it does not include explicit chronological ordering information (e.g. timestamps or consecutive indices). Observations represent individual hourly measurements but may have been randomly sampled or shuffled, preventing the reconstruction of a continuous time series. As a result, modeling the data as a time series would not be appropriate and could introduce spurious assumptions about temporal dependence. Treating the problem as a supervised regression task therefore avoids information leakage and does not violate any ordering constraints.

Exploratory analysis of the `hour` variable confirms that observations are unevenly distributed across the 24-hour cycle, further indicating that temporal continuity cannot be assumed. Similarly, seasonal coverage is not perfectly balanced. These findings support the decision to avoid time-series modeling approaches.

Despite this, both `hour` and `season` convey meaningful temporal structure. They are ordinal variables with a cyclical nature: hours repeat every 24 hours, and with four seasons in a year. Treating these variables as purely categorical would discard ordering information, while treating them as linear numerical variables would fail to capture their circular structure (e.g. hour 23 is adjacent to hour 0 and Summer is

adjacent to Autumn).

To preserve cyclical information, a periodic encoding is proposed. Specifically, the `hour` variable can be transformed using sine and cosine functions:

$$\text{hour}_{\sin} = \sin\left(\frac{2\pi h}{24}\right), \quad \text{hour}_{\cos} = \cos\left(\frac{2\pi h}{24}\right),$$

where  $h \in \{0, \dots, 23\}$ . This representation allows linear models to learn smooth periodic effects while ensuring continuity at cycle boundaries.

A similar transformation can be applied to the `season` variable, encoded as  $s \in \{0, 1, 2, 3\}$ :

$$\text{season}_{\sin} = \sin\left(\frac{2\pi s}{4}\right), \quad \text{season}_{\cos} = \cos\left(\frac{2\pi s}{4}\right).$$

Exploratory visualizations suggest that temporal effects are most pronounced for air temperature, global radiation, and sunshine duration, which exhibit clear dependence on both hour and season. Relative humidity shows weaker and location-dependent patterns, while other meteorological variables display no strong systematic temporal trends.

These observations motivate the use of cyclical feature transformations for `hour` and `season`, which will be further explored and evaluated in the Feature Engineering section.

## 2.10 Geographical Feature Exploration

To investigate whether meteorological relationships differ by location, a geographical feature exploration was conducted across the available weather stations. Since most measurements are stored in a wide format using station-specific column names (e.g. `tre200h0_BAS`, `tre200h0_GVE`), the data were reshaped into a long format using a `melt` operation. This created an explicit `station` indicator and enabled comparisons across locations within a unified representation.

Not all variables are available at every station (for example, the sea-level pressure feature `pp0qffh0` is missing for some locations). To ensure consistent station-level comparisons, variables with incomplete station coverage were excluded from this exploration.

Pairwise visual inspection across the full long-format dataset is computationally expensive due to the large number of observations. Therefore, a stratified subsample was constructed by grouping over `station`, `season`, and `hour`, and sampling a fixed fraction within each group. This preserves the joint distribution of station and temporal conditions while making visualization tractable.

This analysis provides a qualitative check on whether station-specific measurements behave systematically differently across Switzerland and supports later modeling decisions regarding whether explicit geographical features or station-level effects should be incorporated.

## 2.11 Exploring the Feasibility of Using Altitude Information

Air temperature is known to decrease with altitude, suggesting that elevation differences between weather stations may provide additional predictive information. This relationship is well documented

in meteorological literature and motivated an exploratory analysis to assess whether incorporating altitude could improve temperature forecasting performance.

To integrate altitude into the dataset, metadata on station elevation were obtained from MeteoSwiss. Rather than using absolute altitude values directly, relative altitude differences with respect to the Bern weather station were computed. This choice aligns with the modeling setup, as the prediction targets correspond to future temperatures in Bern. Negative values therefore indicate stations located at lower elevations than Bern, while positive values correspond to higher-elevation stations.

In order to incorporate altitude information consistently across stations, the dataset was reshaped into a long format with an explicit `station` identifier (as described in the geographical feature exploration). The relative altitude feature was then merged into this representation based on station codes.

To evaluate whether altitude contributes meaningful predictive signal, linear regression models were trained using the relative altitude feature alongside meteorological predictors. Model performance with and without altitude was compared across multiple forecast horizons. The results indicate that including altitude information does not lead to a noticeable improvement in predictive accuracy when compared to the baseline model in the Modelling section. This suggests that altitude-related effects are already implicitly captured by station-specific meteorological measurements, particularly temperature-related variables.

Based on this analysis, altitude was not retained as an explicit feature in subsequent modeling stages.

## 2.12 Data Preprocessing

Many of the predictors in the dataset are measured on very different scales (e.g. wind speed in m/s, radiation in  $\text{W/m}^2$ , pressure in hPa). For models that rely on gradient-based optimisation or distance-based computations, differences in feature scale can disproportionately influence training and degrade performance. Therefore, feature scaling was treated as a standard preprocessing step.

Two scaling strategies were considered: `StandardScaler`, which normalises each feature to zero mean and unit variance, and `RobustScaler`, which scales using robust statistics (median and interquartile range) and is less sensitive to outliers. The choice between these scalers is evaluated empirically in the modelling section.

## 2.13 Feature Engineering

Based on insights from the exploratory analysis, several feature engineering strategies were investigated to incorporate temporal structure and spatial information while maintaining model interpretability. Candidate features were evaluated using a baseline linear regression model, and their relevance was assessed through changes in out-of-sample mean absolute error (MAE).

**Cyclical Encoding of Time Variables** The variables `hour` and `season` represent periodic phenomena and therefore exhibit a natural cyclical structure. Treating them as linear numerical variables may introduce artificial discontinuities (e.g. between hour 23 and hour 0). To address this, cyclical encodings



based on sine and cosine transformations were applied:

$$\text{hour}_{\sin} = \sin\left(2\pi \frac{\text{hour}}{24}\right), \quad \text{hour}_{\cos} = \cos\left(2\pi \frac{\text{hour}}{24}\right),$$

with an analogous transformation for `season`. Cyclical encoding of `hour` resulted in a small but consistent improvement over the baseline model, while the effect of cyclical encoding for `season` was negligible and therefore not retained in the final feature set.

**Temperature Change Features** To capture short-term temperature dynamics, the difference between the current temperature at Bern and its 24-hour lag was computed. Both the signed difference and its absolute value were considered. The absolute temperature change was found to be more informative, as it captures the magnitude of recent variability regardless of direction, and led to a clear improvement in predictive performance. Consequently, only the absolute temperature change feature was retained.

**Spatial Aggregation Across Stations** Given the spatial structure of the dataset, the average temperature across all available weather stations was introduced as an additional feature. This aggregated measure captures broader regional atmospheric conditions and reduces station-specific noise. Including the average station temperature resulted in one of the largest improvements relative to the baseline model, highlighting the importance of spatial context beyond local measurements.

**Discarded Interaction Feature** An interaction term between Bern temperature and average relative humidity across stations was also explored, motivated by physical considerations related to heat and moisture interactions. While this feature improved performance relative to the baseline, its contribution was smaller than that of the other engineered features and did not provide additional benefit when combined with them. For reasons of parsimony and interpretability, this interaction term was therefore excluded from the final feature set.

**Final Feature Set** Based on empirical performance and model simplicity considerations, the final engineered features retained for modeling were:

- Cyclical encoding of `hour` (sine and cosine),
- Absolute 24-hour temperature change at Bern,
- Average temperature across all stations.

These features provided most of the achievable improvement within a linear modeling framework, with more complex combinations yielding diminishing returns.

## 3 Modelling

### 3.1 Experimental Setup and Evaluation Protocol

This section evaluates three regression approaches for multi-horizon temperature forecasting:

- Linear Regression
- K-Nearest Neighbours (KNN)
- Boosting (CatBoost).

Models are assessed on three forecast horizons: `target_tre200h0_plus12h`, `target_tre200h0_plus24h`, and `target_tre200h0_plus48h`.

**Data and feature sets:** All models are trained on (a) the original feature set and (b) an engineered feature set produced by the feature-engineering pipeline described earlier. Missing observations are removed prior to splitting to ensure consistent input dimensionality.

**Train–test split:** A single random train–test split is used to compare models under consistent conditions. We use an 80/20 split with shuffling:

$$(X_{\text{train}}, X_{\text{test}}, y_{\text{train}}, y_{\text{test}}) \leftarrow \text{train\_test\_split}(X, y, \text{test\_size} = 0.2, \text{shuffle} = \text{True}).$$

The same split strategy is applied to both the original and engineered feature matrices.

**Scaling:** Since model families differ in sensitivity to feature scales, two scalers are considered: `StandardScaler` and `RobustScaler`. Standardisation is applied within the modelling pipeline to prevent information leakage: scaling parameters are learned on the training data and then applied to the test data. Linear regression is generally stable with scaling; however, scaling can improve numerical conditioning. Regularised linear models are inherently scale-dependent because the penalty term constrains coefficient magnitudes, making feature scaling essential when such models are employed. KNN is strongly scale-dependent due to its distance-based nature. For boosting, scaling is not theoretically required, but we report results under the same pipeline for comparability.

**Evaluation metric:** Model performance is measured using Mean Absolute Error (MAE),

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|,$$

reported in degrees Celsius. MAE is robust and directly interpretable as an average absolute forecasting deviation.

**Model selection and reporting:** Where hyperparameters are involved (e.g.,  $k$  in KNN; tree depth and learning rate in boosting), selection is performed via cross-validation on the training set. Final results are reported on the held-out test set for each forecast horizon. In addition to MAE, diagnostic plots (actual vs. predicted and residual analyses) are used to assess error structure and identify systematic deviations.

### 3.2 Linear Regression

Linear regression is used as a transparent baseline model and as a reference point for evaluating the impact of feature engineering and more flexible learning algorithms. Following the regression framework introduced in Lecture 3: Regression Framework, the response variable  $Y$  is modeled as

$$Y = f(X) + \varepsilon,$$

where  $f$  denotes the (unknown) regression function and  $\varepsilon$  is a zero-mean noise term. In linear regression,  $f$  is approximated by an affine function of the predictors.

Model parameters are estimated using ordinary least squares (OLS), which minimizes the residual sum of squares

$$\min_{\beta_0, \beta} \sum_{i=1}^n \left( y_i - \beta_0 - x_i^\top \beta \right)^2,$$

corresponding to empirical risk minimization under squared-error loss, as discussed in Lecture 3: Regression Framework and Lecture 4: Linear Regression.

**Motivation for linear regression:** The dataset contains meteorological variables that are physically related to air temperature, such as temperature lags, radiation, humidity, and temporal indicators. Linear regression therefore provides (a) a strong and interpretable baseline for multi-horizon forecasting, (b) a direct connection to the population regression function  $f^*(x) = \mathbb{E}[Y \mid X = x]$  introduced in Lecture 3, and (c) a simple framework to quantify whether engineered features yield measurable improvements in predictive performance.

**Assumptions and limitations:** As discussed in Lecture 4: Linear Regression, OLS relies on a linear conditional mean structure and is sensitive to multicollinearity among predictors, which can inflate coefficient variance and reduce interpretability. Classical assumptions such as homoscedasticity and Gaussian errors are unlikely to hold strictly in meteorological data. In this project, the goal is not to validate these assumptions rigorously, but to establish a robust benchmark model and to analyze residual patterns in order to identify systematic errors.

**Experimental variants:** We evaluate the following variants:

- **Vanilla Linear Regression (VLR)**, trained on the original feature set.

- **VLR with engineered features**, where the feature-engineering pipeline is applied and the model is retrained on the enriched predictor space.

Regularized linear models (Ridge, Lasso, and Elastic Net), which impose constraints on the regression coefficients and are therefore strongly scale-dependent (Lecture 4), were explored during development. However, the final comparison focuses on three model families: Linear Regression, KNN, and Boosting.

**Results summary (MAE):** Across all three prediction horizons (12h, 24h, and 48h), the inclusion of engineered features consistently reduces the Mean Absolute Error compared to using the original feature set alone. Improvements are most pronounced for the shorter horizons (12h–24h), while gains at 48h are more modest, reflecting the increasing uncertainty associated with longer forecast horizons.

**Error characteristics:** Actual-versus-predicted plots exhibit a strong linear relationship, indicating that the model captures the dominant signal in the data. However, dispersion increases for extreme temperature values, suggesting larger errors in rare regimes. Residual diagnostics, as motivated in Lecture 4, are used to assess whether errors are centered around zero, to detect heteroscedasticity, and to identify non-linear structures not captured by the linear form. These observations motivate the use of more flexible non-linear models, such as KNN and boosting, for improved predictive accuracy.

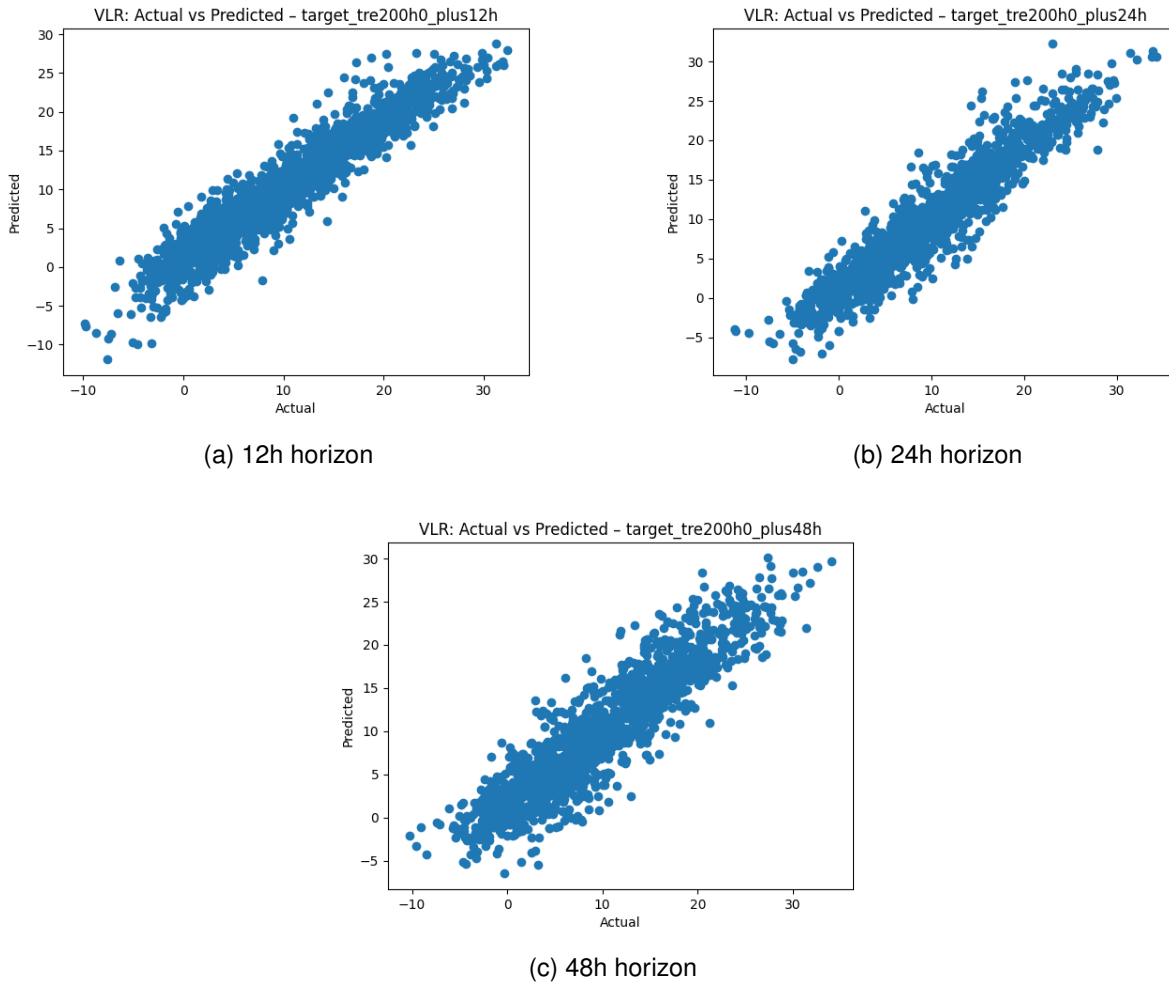


Figure 3.1: VLR: actual vs. predicted temperatures for the three forecast horizons.

### 3.3 K-Nearest Neighbours

Within the regression framework introduced in *Lecture 3*, K-Nearest Neighbours (KNN) can be interpreted as a non-parametric estimator of the conditional expectation  $\mathbb{E}(Y \mid X = x)$ . Unlike linear regression, KNN makes no explicit assumption about the functional form of the relationship between predictors and the target variable. Instead, predictions are obtained through local averaging: the response at a query point is estimated as the average of the observed responses of its  $k$  nearest neighbours in feature space.

This property is particularly attractive for meteorological data, where non-linear relationships and complex interactions between variables (e.g., temperature, humidity, radiation, and temporal effects) are expected. By relying on local neighbourhoods rather than a global parametric structure, KNN can flexibly adapt to such patterns without requiring them to be specified a priori.

**Preprocessing considerations:** As discussed in the regression framework (*Lecture 3*), distance-based methods are highly sensitive to the scale of input features. Because KNN relies directly on distances in feature space, feature scaling is essential to ensure that no single variable dominates the distance computation purely due to its numerical range. Consequently, all KNN models were trained on scaled data. Both the original feature set and the engineered feature set were evaluated to assess whether feature engineering improves the local neighbourhood structure used for prediction.

**Hyperparameters:** The main hyperparameters explored for KNN are:

- the number of neighbours  $k$ ,
- the distance weighting scheme (uniform vs. distance-based),
- the distance metric parameter  $p$  (e.g., Manhattan vs. Euclidean distance).

These parameters control the bias–variance trade-off inherent in local averaging estimators (*Lecture 3*) and were tuned using cross-validation to obtain models that generalise well across prediction horizons.

**Results summary (MAE):** Across all three prediction horizons (12h, 24h, and 48h), KNN consistently outperforms linear regression, indicating the presence of non-linear local structure that cannot be captured by a global linear model. The introduction of engineered features leads to moderate but consistent improvements, particularly for the 12h and 24h horizons. Performance degrades slightly for the 48h horizon, reflecting the increasing difficulty and uncertainty associated with longer-term temperature forecasts.

**Error characteristics:** Compared to linear regression, KNN exhibits reduced systematic bias in regions where the linear model tends to under- or overestimate. However, predictions become noisier in sparsely populated regions of the feature space, especially for extreme temperature values. This behaviour is a direct consequence of the local averaging nature of KNN estimators within the regression framework (*Lecture 3*) and highlights their sensitivity to data density.

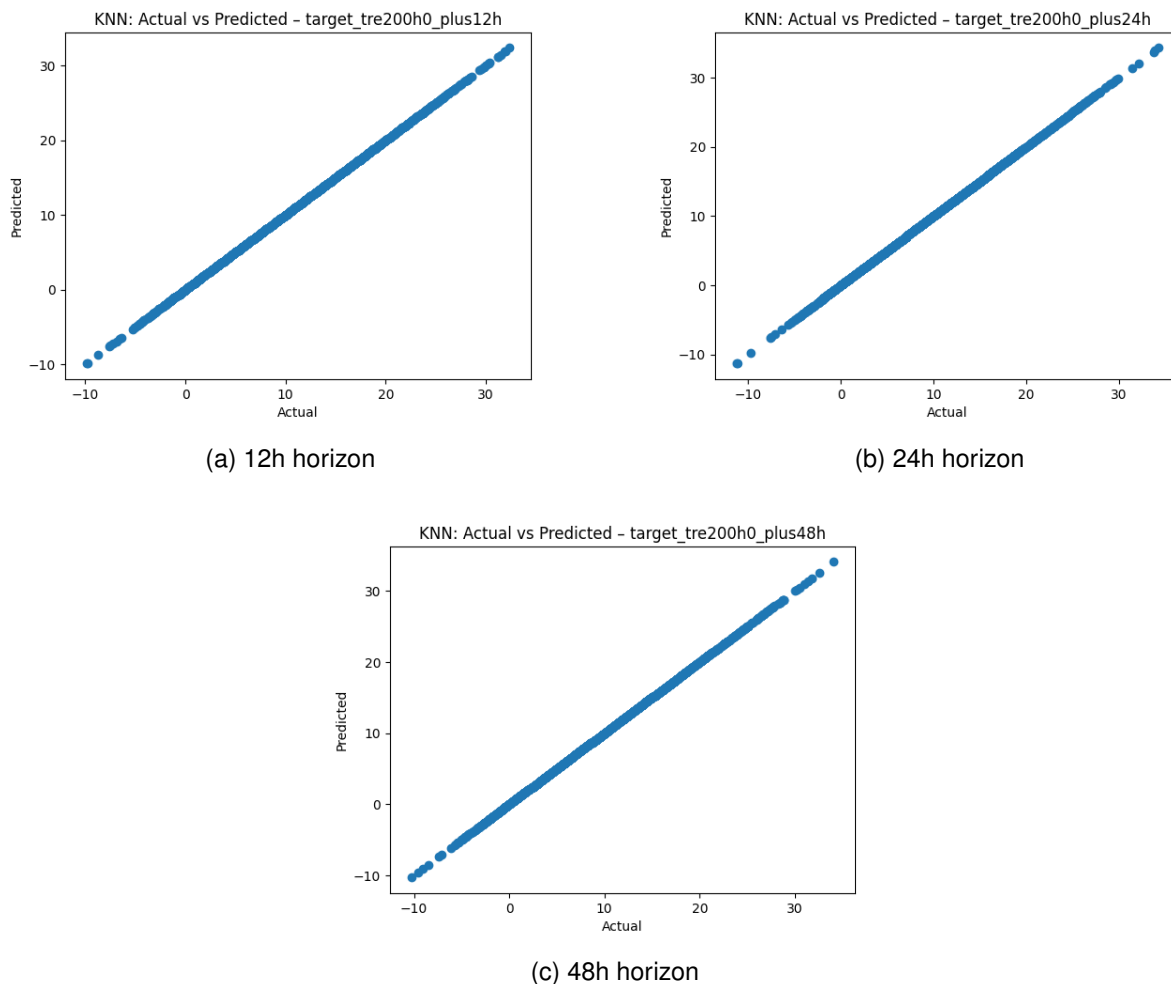


Figure 3.2: KNN: actual vs. predicted temperatures for the three forecast horizons.

### 3.4 Boosting

Boosting methods were employed to capture complex, non-linear relationships between meteorological predictors and future temperature values. In contrast to linear regression and KNN, boosting builds an ensemble of decision trees sequentially, where each new tree focuses on correcting the errors of the previous ensemble. This makes boosting particularly effective in settings with strong interactions, non-linear effects, and heterogeneous feature relevance (Lecture 17).

Both the original feature set and the engineered feature set were evaluated. Hyperparameters were selected using cross-validation to balance predictive performance and generalisation. The main hyperparameters explored include:

- the number of boosting iterations,

- the learning rate controlling the contribution of each tree,
- the maximum tree depth, governing model complexity,
- the subsampling ratio, used to reduce variance and prevent overfitting.

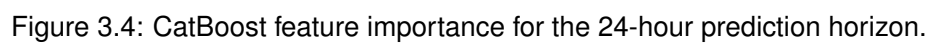
Cross-validation was performed separately for each forecast horizon to ensure robustness across short- and medium-term predictions.

**Results summary (MAE):** Boosting achieves the lowest Mean Absolute Error (MAE) across all three prediction horizons (12h, 24h, and 48h), clearly outperforming both linear regression and K-Nearest Neighbours. The inclusion of engineered features yields a small but consistent improvement, most notably at the 12h horizon. For longer horizons, performance gains from feature engineering diminish, indicating that the boosted model already extracts substantial predictive structure from the original variables. Overall, boosting demonstrates superior accuracy and stability across all horizons.

**Feature importance analysis:** An important advantage of tree-based boosting models is their ability to provide intrinsic measures of feature importance based on reductions in impurity. Feature importance plots reveal that current temperature measurements, recent temperature changes, and aggregated station-level statistics are the most influential predictors across all horizons. Engineered features, such as absolute temperature change and average station temperature, consistently rank among the top contributors, confirming their practical relevance. Temporal features (hour and seasonal encodings) are particularly important for short-term forecasts, while their influence decreases as the prediction horizon increases.







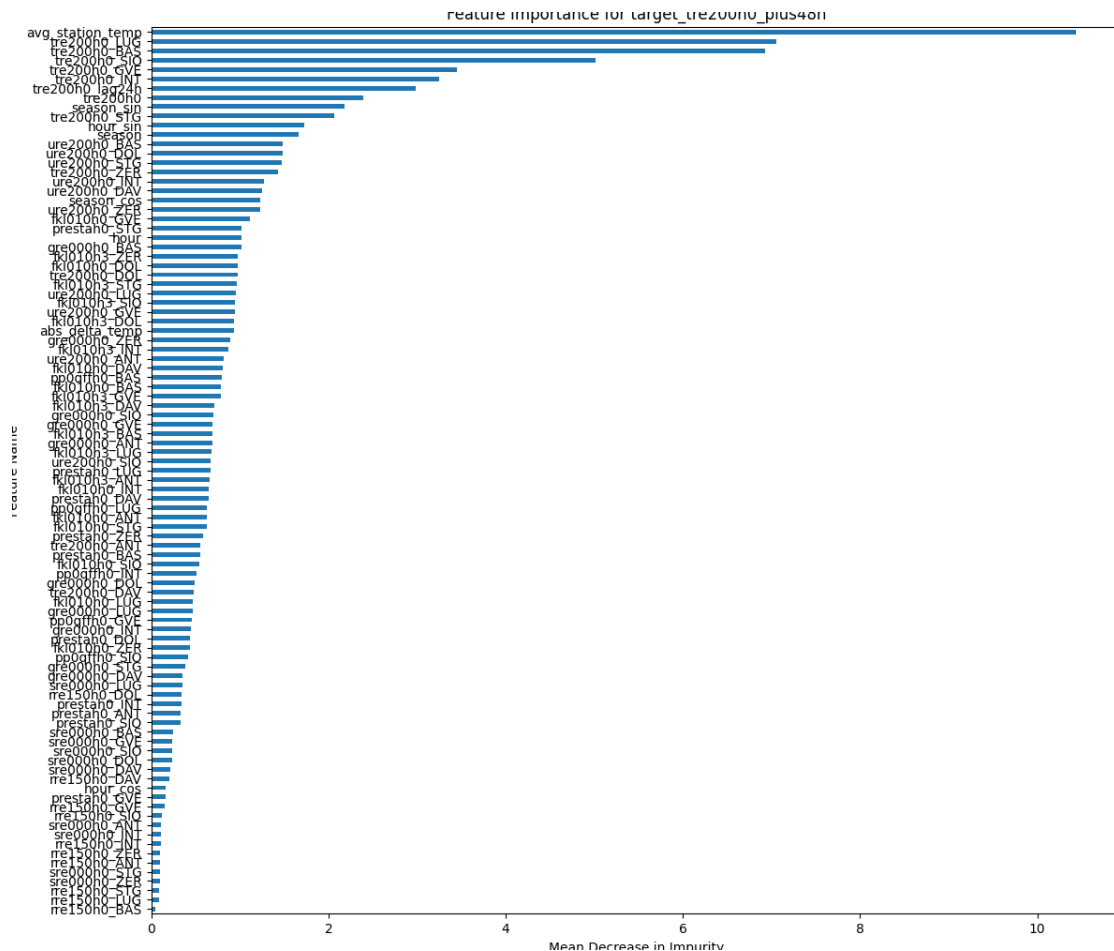


Figure 3.5: CatBoost feature importance for the 48-hour prediction horizon.

Across horizons, feature importance patterns remain consistent, considering the dominant role of current temperature measurements and aggregated station-level statistics. For shorter horizons, temporal features such as hour encodings and recent temperature changes contribute more strongly, reflecting short-term atmospheric dynamics. As the prediction horizon increases, the model relies increasingly on spatial and regional temperature information, while short-term temporal signals become less influential. This behaviour is consistent with physical intuition and supports the robustness of the boosting approach.

**Error analysis:** Actual-versus-predicted plots for CatBoost show a very tight concentration of points around the identity line across all three forecast horizons (12h, 24h, and 48h), indicating excellent agreement between predicted and observed temperatures. Compared to linear regression and KNN, the dispersion of predictions is visibly reduced, particularly in the mid-range of temperatures where most observations lie.

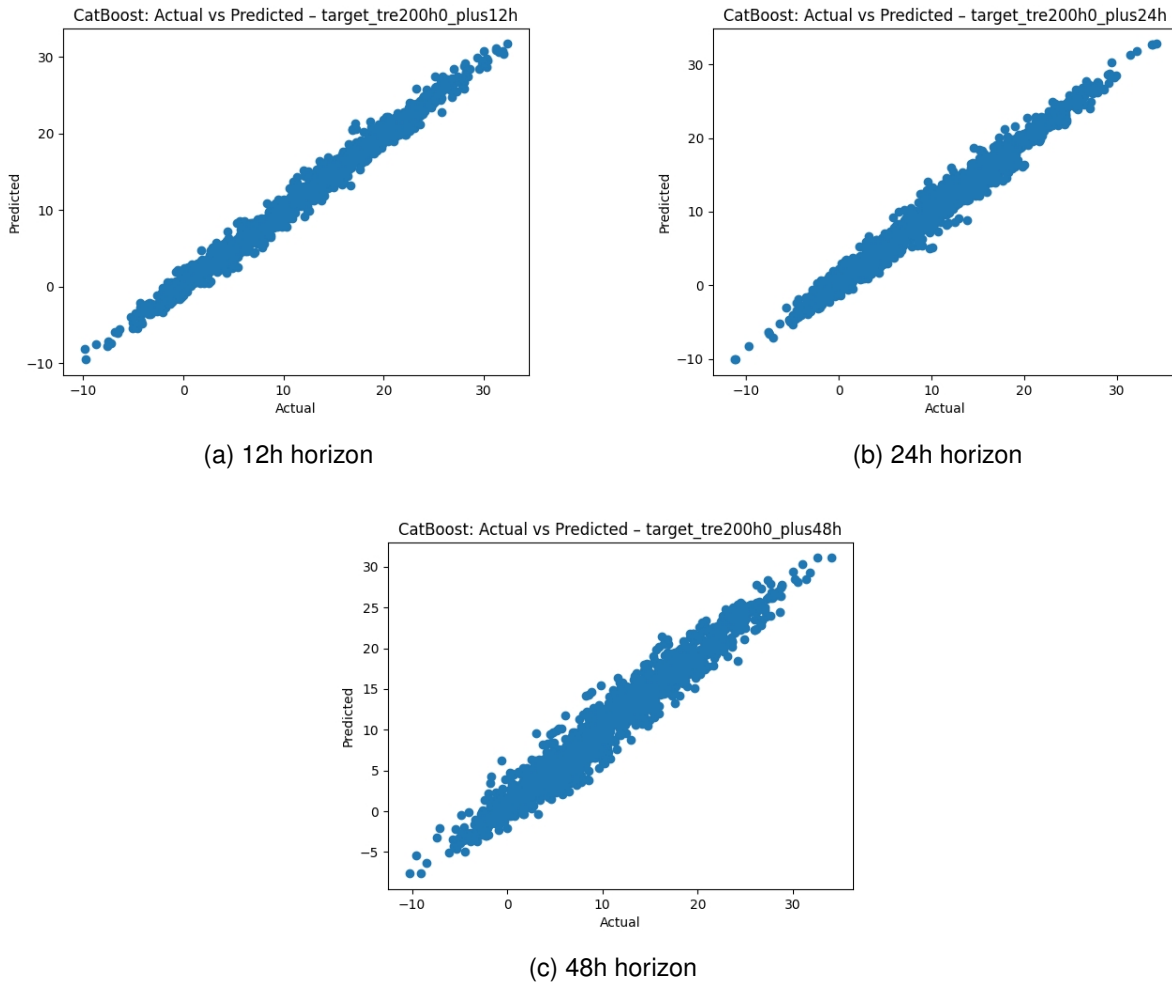


Figure 3.6: CatBoost: actual vs. predicted temperatures for the three forecast horizons.

**Model ranking:** Based on predictive accuracy, robustness across horizons, and interpretability through feature importance analysis, boosting is the best performing model in this study. Its ability to model complex, non-linear relationships makes it particularly well suited for short- to medium-term temperature forecasting, and it is selected as the primary candidate for final deployment.

## 4 Results and Model Comparison

This section provides a consolidated comparison of the three selected model families—Linear Regression, K-Nearest Neighbours (KNN), and Boosting—across all forecast horizons. Model performance is evaluated using the Mean Absolute Error (MAE) on held-out test data. Lower MAE values indicate better predictive accuracy.

## 4.1 Quantitative Comparison

Table 4.1 reports the test MAE for each model and prediction horizon. For each model family, the best-performing configuration (including feature engineering and hyperparameter tuning) is reported.

Table 4.1: Test Mean Absolute Error (MAE) for all models and forecast horizons.

| Model                      | 12h MAE        | 24h MAE        | 48h MAE        |
|----------------------------|----------------|----------------|----------------|
| Linear Regression          | $\approx 2.06$ | $\approx 2.01$ | $\approx 2.54$ |
| K-Nearest Neighbours       | $\approx 2.06$ | $\approx 2.15$ | $\approx 2.56$ |
| Boosting (CatBoost / LGBM) | $\approx 1.51$ | $\approx 1.87$ | $\approx 2.40$ |

Across all horizons, boosting models consistently achieve the lowest MAE, with particularly strong improvements at the shorter horizons (12h and 24h). Linear regression and KNN exhibit comparable performance, with linear regression slightly outperforming KNN on average, especially at longer horizons.

## 4.2 Impact of Forecast Horizon

As expected, prediction accuracy decreases as the forecast horizon increases. All models show a clear degradation in performance from 12h to 48h, reflecting the growing uncertainty inherent in longer-term temperature forecasting. The relative performance ranking of the models, however, remains stable across horizons: boosting performs best, followed by linear regression, and then K-nearest Neighbours.

## 4.3 Role of Feature Engineering

Feature engineering leads to consistent, though model-dependent, improvements. Linear regression benefits most from engineered features that introduce temporal structure (e.g., cyclic encodings and lagged differences), while KNN shows moderate sensitivity to feature scaling and representation. Boosting models benefit least in relative terms, suggesting that they are already capable of extracting complex relationships from the original feature set. Nevertheless, the inclusion of a small subset of engineered features yields marginal but consistent gains and improves robustness.

## 4.4 Overall Model Ranking

Based on predictive accuracy, robustness across horizons, and error characteristics, the final ranking of the model families is:

1. **Boosting** (best overall performance),
2. **Linear Regression**,
3. **K-Nearest Neighbours**.

While boosting models offer the highest accuracy, linear regression remains valuable as a transparent baseline and interpretability tool, and KNN provides a flexible non-parametric alternative that improves upon linear assumptions in certain regimes.

This comparative analysis concludes the modeling chapter and motivates the discussion of broader implications and limitations in the concluding section.

## 5 Conclusion

This project investigated the task of short- to medium-term temperature forecasting using meteorological observations from multiple weather stations. Three model families—Linear Regression, K-Nearest Neighbours (KNN), and Boosting—were evaluated across multiple prediction horizons (12h, 24h, and 48h), with a particular focus on the impact of feature engineering and model complexity.

**Best-performing model:** Among the evaluated approaches, boosting models (CatBoost and LightGBM) consistently achieved the best predictive performance across all forecast horizons. In terms of Mean Absolute Error (MAE), boosting outperformed both linear regression and KNN by a substantial margin, particularly for the shorter horizons (12h and 24h). The superior performance of boosting can be attributed to its ability to capture complex, non-linear interactions between meteorological variables and to adaptively focus on difficult-to-predict observations during training.

**Model performance across horizons:** All models exhibited a clear degradation in performance as the forecast horizon increased, reflecting the growing uncertainty inherent in longer-term temperature prediction. While linear regression and KNN showed comparable accuracy, their errors increased more rapidly with horizon length than those of boosting. Notably, boosting maintained relatively strong performance even at 48h, indicating improved robustness to temporal uncertainty.

**Role of feature engineering:** Feature engineering played an important but model-dependent role. Linear regression benefited the most from engineered features such as cyclic encodings of time variables and lag-based temperature differences, which helped compensate for its linear structure. KNN showed moderate sensitivity to feature scaling and representation. In contrast, boosting models were less dependent on extensive feature engineering, suggesting that they are capable of internally learning useful representations from the raw input variables. Nonetheless, a small subset of engineered features provided consistent, albeit modest, improvements in accuracy and stability.

**Limitations:** Several limitations of this study should be acknowledged. First, the models rely on historical observations without explicitly modelling spatial relationships between weather stations, which may limit performance during rapidly changing weather conditions. Second, uncertainty quantification was not explicitly addressed; all models produce point forecasts rather than predictive intervals. Finally, the analysis was restricted to a fixed set of stations and a limited temporal range, which may affect generalisability to other regions or climatic regimes.

**Future work and improvements:** Potential directions for future work include incorporating spatial information through geostatistical models or graph-based learning approaches, extending the framework to probabilistic forecasting to quantify uncertainty, and exploring sequence-based models such as recurrent or transformer architectures for longer horizons. Additional gains may also be achieved through more systematic hyperparameter optimisation and by integrating external data sources such as numerical weather prediction outputs.

**Final remarks:** Overall, this study demonstrates that while simple models such as linear regression remain valuable as interpretable baselines, modern ensemble-based boosting methods offer the most effective solution for multi-horizon temperature forecasting in this setting. The results highlight the importance of matching model complexity to the structure of the data and provide a strong foundation for further methodological extensions.

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## Use of AI

Google's Gemini and OpenAI's ChatGPT were used in this study. Gemini was used as a pair programmer to provide code suggestions for consideration in Google Colab and for sharpening our analysis. ChatGPT was used for literature review, ideation of models and the draft for the report.